

Applying the Galerkin Method

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Abstract

We develop an understanding of the Galerkin Method that allows us to solve nonlinear ODEs and time dependent PDEs. Considering only continuous basis functions, we develop several related python codes that solve the select problems outlined here. These codes serve as an outline that could be used to create a more robust code in the future. We find that our results agree well with analytic solutions or other numerical solutions, where available. When unavailable, we rely on qualitative physical understandings of problems, which also show agreement.

1 Intro

To understand the Galerkin method, we must first grasp the concept of a weak form differential equation. Following in the footsteps of Boyd [1], suppose that we have some differential operator, $\hat{L} = \partial_{xx} + V(x)$, and a forcing function, f , to go along with it: $\hat{L}u = f$. We are going to solve this PDE on a domain (a, b) . We assume, for now, that this problem has the simple boundary conditions $u(a) = u(b) = 0$.

Aside: We define the inner (scalar) product as

$$\langle f, g \rangle = \int_a^b f(x)g(x)dx$$

We can now choose a test function $\varphi(x)$, which also satisfies the boundary conditions, and take its inner product with the PDE:

$$\begin{aligned}\langle \varphi, \hat{L}u \rangle &= \langle \varphi, f \rangle \\ \langle \varphi, u_{xx} \rangle + \langle \varphi, Vu \rangle &= \langle \varphi, f \rangle \\ \varphi u_x \Big|_a^b - \langle \varphi_x, u_x \rangle + \langle \varphi, Vu \rangle &= \langle \varphi, f \rangle \\ \langle \varphi, Vu \rangle - \langle \varphi_x, u_x \rangle &= \langle \varphi, f \rangle\end{aligned}\tag{1}$$

We use integration-by-parts to greatly simplify our new differential equation, also taking advantage of the Dirichlet condition to cancel boundary terms.

This new differential equation is the “weak-form” expression of our original problem. This form has two major benefits. First, any function with an integrable first derivative could satisfy this equation, which widens both the options for test functions φ and solutions u . This is especially useful in fluid mechanics where there’s a need to model shock waves, which do not have a continuous first derivative. Second, we only need to calculate first derivatives for use in our numerical scheme, which means that coordinate transformations can be done with greater ease.

2 Galerkin’s Method

The Galerkin Method fits in a broader set of finite element methods, called Mean Weighted Residual Methods, which generally work in the following way:

- Pick a set of basis functions (φ_n) that satisfy the appropriate boundary conditions.
- Pick a suitable set of test functions.
- Calculate the residual of the system. For an operator $\hat{L}$, and approximation $u_N = \sum_n c_n \varphi_n(x)$, the residual is $R = \hat{L}u_N - f$, where f is the inhomogeneous part of the differential equation.
- Minimize the inner product of the residual with the test functions.

Of interest to us is the Galerkin Method, which uses the same set of functions (φ_n) as both basis and test function. The residual then has the following form:

$$\begin{aligned}\langle \varphi_m, R \rangle &= \langle \varphi_m, \hat{L} \left[\sum_n c_n \varphi_n(x) \right] - f \rangle \\ &= \sum_n \left[c_n \langle \varphi_m, \hat{L} \varphi_n \rangle \right] - \langle \varphi_m, f \rangle\end{aligned}$$

To minimize the residual, we enforce the condition that $\langle \varphi_m, R \rangle = 0$ for each value of m :

$$\sum_n \left[c_n \langle \varphi_m, \hat{L} \varphi_n \rangle \right] = \langle \varphi_m, f \rangle$$

Breaking into the individual ms and ns , we see that

$$\begin{aligned}\langle \varphi_0, f \rangle &= c_0 \langle \varphi_0, \hat{L} \varphi_0 \rangle + c_1 \langle \varphi_0, \hat{L} \varphi_1 \rangle + \dots + c_n \langle \varphi_0, \hat{L} \varphi_n \rangle \\ \langle \varphi_1, f \rangle &= c_0 \langle \varphi_1, \hat{L} \varphi_0 \rangle + c_1 \langle \varphi_1, \hat{L} \varphi_1 \rangle + \dots + c_n \langle \varphi_1, \hat{L} \varphi_n \rangle \\ &\vdots \\ \langle \varphi_m, f \rangle &= c_0 \langle \varphi_m, \hat{L} \varphi_0 \rangle + c_1 \langle \varphi_m, \hat{L} \varphi_1 \rangle + \dots + c_n \langle \varphi_m, \hat{L} \varphi_n \rangle.\end{aligned}$$

Note that this has the form of a matrix equation

$$\vec{F} = \mathbf{L}\vec{c}, \tag{2}$$

where $F_m = \langle \varphi_m, f \rangle$, $L_{mn} = \langle \varphi_m, \hat{L}\varphi_n \rangle$, and $\vec{c} = c_n$. There is one obvious important thing to note here: $\mathbf{L}$ will always be invertible for a set of orthogonal polynomials (see the first proof in the appendix).

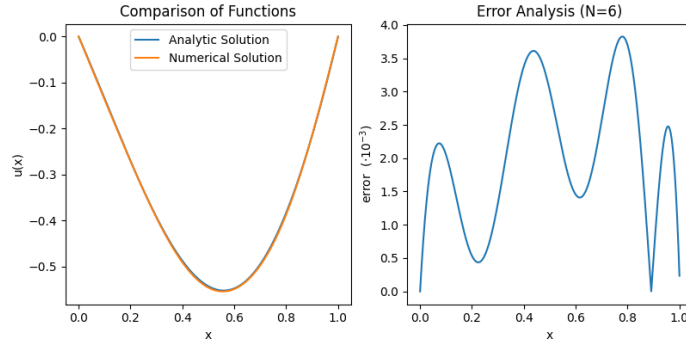
A second and less obvious fact comes from examination of Equation 1. Any operator $\hat{L}$ which can be reduced to a weak-form differential equation where φ is symmetrically differentiated will have a symmetric matrix $\mathbf{L}$ (see the second proof in the appendix). This is highly valuable since there are special methods for inverting symmetric matrices, which are more computationally efficient than their non-symmetric counterparts.

2.1 Examples

Consider $u'' - u' + 9u = x$ with boundary conditions $u(0) = u(1) = 0$, which has the analytic solution

$$u(x) = \frac{1}{81} \left(9x + e^{\frac{x-1}{2}} \csc\left(\frac{\sqrt{35}}{2}\right) \left(\sqrt{e} \sin\left(\frac{1}{2}\sqrt{35}(x-1)\right) - 10 \sin\left(\frac{\sqrt{35}x}{2}\right) \right) + 1 \right)$$

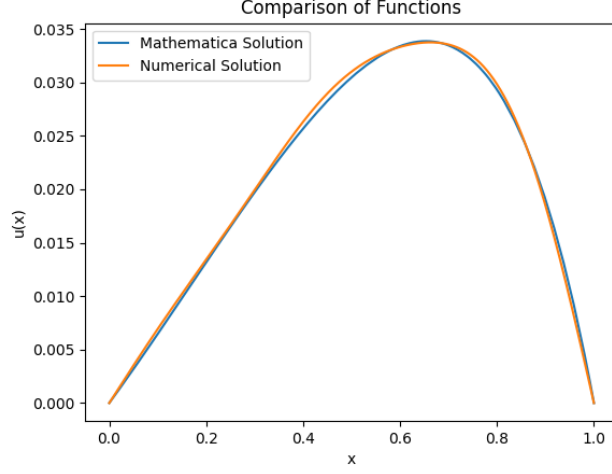
Using the Galerkin Method with $\varphi_n(x) = \sin\left(\frac{n\pi}{a}x\right)$, $a = 1$ and $N = 6$, we find the numerical solution, which agrees well with the analytic solution:



Having gained some confidence, we examine a non-linear problem (with the same boundary conditions and basis):

$$u'' + \frac{u'}{u^2 + 1} - 9u = -x$$

We find that this agrees quite well with Mathematica's built in NDSolve function:



3 Introducing Time-Dependence

Up to this point, we have essentially developed a method that solves any ordinary differential equation, given a suitable set of basis functions. But the world is described by partial differential equations, and almost all of them have some sort of time dependence. Let's consider the (dimensionless) diffusion equation:

$$u_t - u_{xx} = f(x, t) \quad (3)$$

with Dirichlet boundary condition $u(0, t) = u(a, t) = 0$ and $u(x, 0) = u_0(x)$. For this problem, we choose u such that $u_N = \sum_n^N c_n(t) \varphi_n(x)$. Plugging this solution into Equation 3, we see the following

$$f(x, t) = \sum_n^N c_{n,t}(t) \varphi_n(x) - \sum_n^N c_n(t) \varphi_{n,xx}(x)$$

$$\langle \varphi_m, f \rangle = \langle \varphi_m, \sum_n^N c_{n,t}(t) \varphi_n(x) \rangle - \langle \varphi_m, \sum_n^N c_n(t) \varphi_{n,xx}(x) \rangle.$$

Working through the algebra and integrating by parts, we have

$$\begin{aligned} \langle \varphi_m, f \rangle &= \langle \varphi_m, c_{n,t}(t) \varphi_n(x) \rangle - \langle \varphi_m, c_n(t) \varphi_{n,xx}(x) \rangle \\ \langle \varphi_m, f \rangle &= c_{n,t}(t) \langle \varphi_m, \varphi_n(x) \rangle - c_n(t) \langle \varphi_m, \varphi_{n,xx}(x) \rangle \\ \langle \varphi_m, f \rangle &= c_{n,t}(t) \langle \varphi_m, \varphi_n(x) \rangle - \varphi_m \varphi_{n,x}(x) \Big|_0^a + c_n(t) \langle \varphi_{m,x}(x), \varphi_{n,x}(x) \rangle \\ \langle \varphi_m, f \rangle &= c_{n,t}(t) \langle \varphi_m, \varphi_n(x) \rangle + c_n(t) \langle \varphi_{m,x}, \varphi_{n,x}(x) \rangle \end{aligned}$$

which is the matrix equation

$$\vec{F} = \mathbf{M} \vec{c}_t + \mathbf{L} \vec{c}$$

where $F_m = \langle \varphi_m, f \rangle$, $L_{nm} = \langle \varphi_m, x, \varphi_n, x \rangle$, $M_{nm} = \langle \varphi_m, \varphi_n \rangle$, $\vec{c} = c_m(t)$, and $\vec{c}_t = \frac{d\vec{c}}{dt}$. That's great, but to solve this equation in time, we must put it in a more suggestive form. Following Demlow [3], we solve for $\vec{c}_t$ to get

$$\vec{c}_t = \mathbf{M}^{-1}(\vec{F} - \mathbf{L}\vec{c}) \quad (4)$$

and can choose our favorite method to solve a vector ODE (for this paper, fourth order Runge-Kutta).

3.1 Example

The form of Equation 3 is quite similar to that of the Time-Dependent Schrödinger Equation (TDSE) in one dimension:

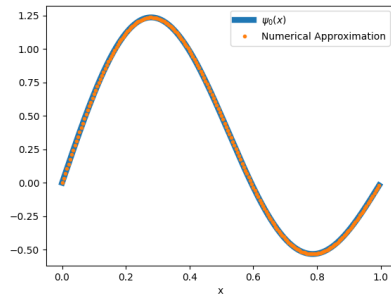
$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V(x)\Psi(x, t) = i\hbar \frac{\partial \Psi}{\partial t}, \quad (5)$$

For simplicity, we analyze the Infinite Square Well potential

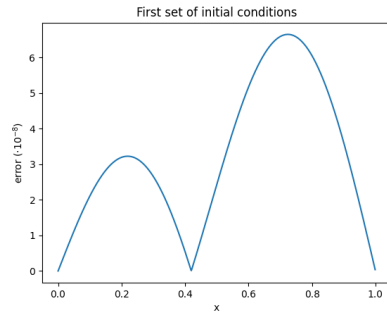
$$V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & \text{otherwise} \end{cases}$$

with a Dirichlet boundary condition (informed by Quantum Mechanics) and $a = 1$. Again, we will use sines as the basis and assume that $\Psi(x, 0) = \psi_0(x)$. To begin the simulation, we must find $\vec{c}_0$, which are the Fourier coefficients (c_n) at $t = 0$. Then, we put the TDSE into a form more similar to Equation 4 and evolve in time.

For $\psi_0(x) = \frac{1}{2}\varphi_1 + \frac{\sqrt{3}}{2}\varphi_2$, we expect to see the solution modeled exactly, “sloshing” back and forth between the two states above.



(a) $t = 0$



(b) Error

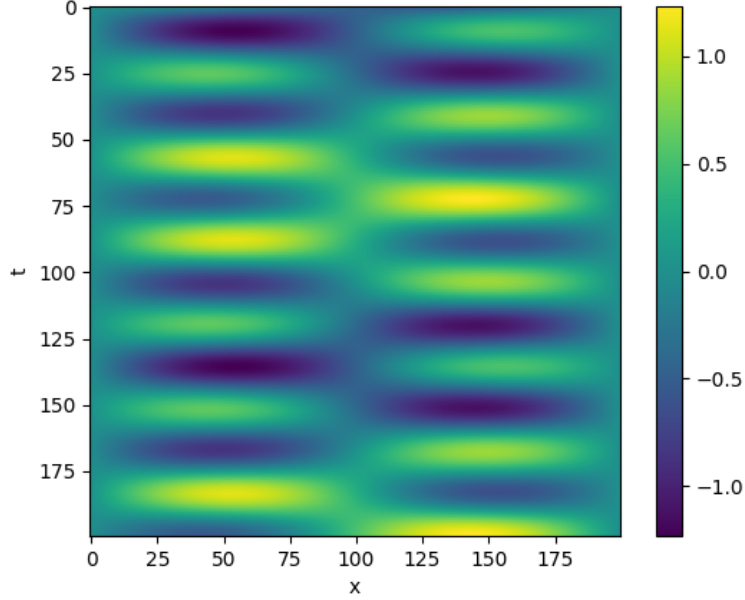


Figure 2: Evolving in time

We see excellent agreement for the initial conditions and expect the time evolution to be quite accurate as well. When testing, we found that increasing the number of terms did change the initial behavior. Further, we found that decreasing our step size parameter for initial condition integration did significantly decrease the error, but not to the level that Simpson's rule (a $\mathcal{O}(h^5)$ approximation) should. For a step size on the order of 10^{-2} , we see error on the order of 10^{-8} .

So does this agree? According to the Composite Simpson's rule (Burden and Faires)

$$\text{error} \left[\int_a^b \left(\frac{1}{2} \sin(\pi x) + \frac{\sqrt{3}}{2} \sin(2\pi x) \right) \sin(n\pi x) dx \right] \leq \max \left(\frac{d^4 f(\xi)}{dx^4} \right) \frac{h^4}{180} (b-a)$$

where $f(\xi) = \psi_0(\xi) \sin(n\pi\xi)$ is determined for each n we use to calculate a weight. To find the maximum value of $\frac{d^4 f}{dx^4}$, we must calculate $\frac{d^5 f}{dx^5}$ and determine its root. This is a perfect opportunity for numerical root-finding. We start by looking at the function when $n = 1$:

$$f^{(IV)}(x) = \frac{1}{4} \pi^4 \left(\sqrt{3} \cos(\pi x) - 16 \cos(2\pi x) - 81\sqrt{3} \cos(3\pi x) \right)$$

which has its maximum at the point where

$$f^{(V)}(x) = \frac{1}{2}\pi^5 \sin(\pi x) \left(32 \cos(\pi x) + \sqrt{3}(243 \cos(2\pi x) + 121) \right)$$

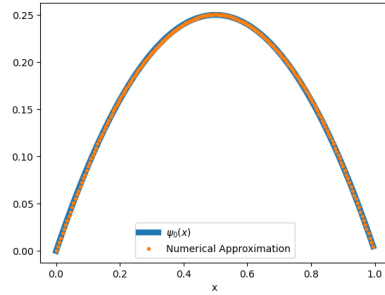
has a root. To find this root, we use Newton's Method

$$x_{n+1} = x_n - \frac{f^{(V)}(x_n)}{f^{(VI)}(x_n)}, \quad n = 0, 1, \dots$$

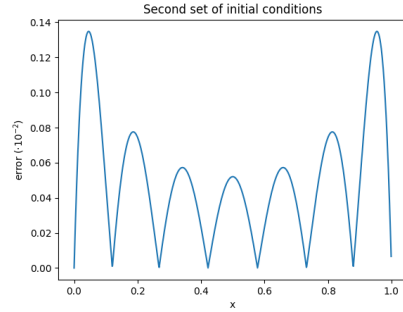
$$f^{(VI)}(x) = \frac{1}{4}\pi^6 \left(-\sqrt{3} \cos(\pi x) + 64 \cos(2\pi x) + 729\sqrt{3} \cos(3\pi x) \right)$$

with a tolerance of 10^{-5} to find that the maximum value $f^{(IV)} \approx 3638.8932815$ is at $x \approx 0.3397721$. Repeating this same process for the case where $n = 2$, we find a maximum value $f^{(IV)} \approx 10899.9007352$ at $x \approx 0.5109241$. We find the total error bound by summing the individual bounds. The total value of $8.07710737 \cdot 10^{-7}$ is indeed greater than the error we see from the graph.

We now consider $\psi_0(x) = -(x - \frac{a}{2})^2 + \frac{a^2}{4}$ and watch initial behavior and time evolution for $N = 6$.



(a) $t = 0$



(b) Error

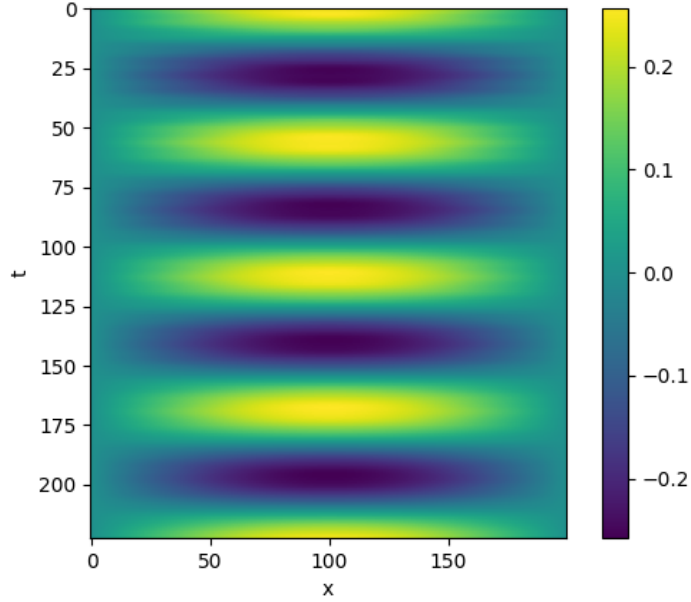


Figure 4: Evolving in time

Again, we see agreement for the initial conditions, and expect high accuracy for the time dependent behavior. This is also a valuable time to notice that this scheme works as well for real valued differential equations as complex valued ones, which makes it very powerful.

4 Extending to higher order (in time)

Successfully creating a scheme that works for first time derivatives is great, but if we could somehow extend this to second time derivatives, we would have a starting point for modeling basically any physical phenomenon. Thankfully, there exists a simple trick we can use to extend our time scheme: variable substitution. Consider the non-dimensional wave equation:

$$u_{tt} - u_{xx} = f(x, t, u_t)$$

Using our current scheme, this second order in time equation is difficult to work with and this is exacerbated by the source term. So, we make the substitution

$v = u_t$, giving the following system

$$\begin{aligned}\frac{dv}{dt} &= u_{xx} + f(x, t, v) \\ \frac{du}{dt} &= v\end{aligned}$$

We have thus converted the second order equation into two first order equations. This may seem entirely unhelpful since we still need to solve this system of equations. However, creating a vector $\vec{U} = (u, v)^T$, we can redefine the system as

$$\frac{d\vec{U}}{dt} = \vec{F}(\vec{U}, t) \quad (6)$$

where $\vec{F}$ is an appropriate vector-valued function. Note that this has the same form as Equation 4, so we can use the fourth order Runge-Kutta method here as well.

But we want to extend this concept to the Galerkin Method. For this to work, we consider the weights c_n to be time-dependent, and construct the vector

$$\vec{C} = \begin{pmatrix} c_1 \\ \vdots \\ c_N \\ \frac{dc_1}{dt} \\ \vdots \\ \frac{dc_N}{dt} \end{pmatrix}$$

Doing the analogous algebra that led to Equation 6, we find that

$$\frac{d\vec{C}}{dt} = \vec{G}(\vec{C}, t) \quad (7)$$

where $\vec{G}$ is a vector-valued ‘‘Galerkin Function’’ that describes relationships between the weights and their first and second time derivatives.

4.1 Example

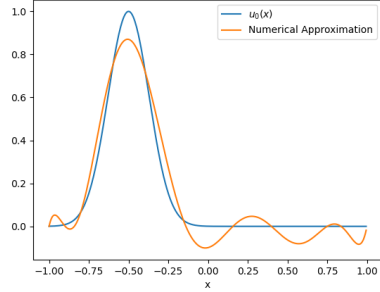
Consider the (1 + 1)D Wave Equation on a symmetric domain $(-a, a)$ with Dirichlet boundary condition and a specified initial configuration $u_0(x)$

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0 \quad (8)$$

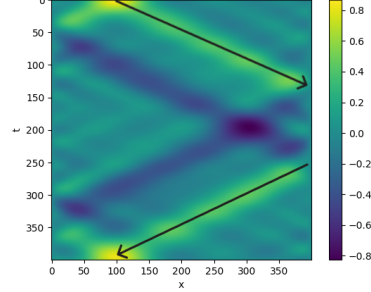
where we choose units such that $c = 1$.

Letting $u(x, t) = \sum_n^N c_n(t) \varphi_n(x)$, we can put this in the form of Equation 7, and can now use the fourth order Runge-Kutta method to solve the system.

As an example, we choose $u_0(x) = e^{-30x^2}$ (for more information on the initial conditions, see the second appendix) and use homogeneous Chebyshev Polynomials (up to $N = 9$) to model the time evolution.



(a) $t = 0$



(b) Evolving in time

Notice that for the time-evolution, we see propagation of information (represented by the black arrows) at a constant speed. Further, we see a reflection of the incident wave off the boundary that flips the wave (the yellow incident wave flips and travels back as the purple reflected wave). These are both important characteristics of the Wave Equation and suggest that our model is accurate.

Conclusion

Now that we have code that solves each problem we laid out, we can extend this to virtually any (1+1)D PDE with minimal changes. A first improvement would be to refine the code to make use of more pythonic, time-efficient structures, or to simply switch the code to a more efficient language (like C).

For the future, it would be valuable to extend this framework to multiple dimensions, still with time-dependent behavior. A good test for this would be setting up a solver to model the full 3D time-dependent hydrogen atom (a classic problem in physics). With considerations into higher dimensions, the idea of a Discontinuous Galerkin method arises. Using piecewise linear functions as the basis, one can extend the techniques from this paper to irregularly shaped higher dimensional domains (see Hesthaven and Warburton [5]). For now though, the solver we have developed works for a wide range of problems and is a good starting point for any further work.

Appendices

Homogeneous Chebyshev Polynomials

Following Boyd’s “Bestiary of Basis Functions” (pg. 497) we define the Chebyshev Polynomials of the first kind as follows:

$$T_n(\theta) = \cos(n\theta)$$

Essentially, these are trigonometric functions in disguise. We then know that the Chebyshev series should take a form reminiscent of a Fourier Series. Specifically, Dobrushkin [4] lays out the form as:

$$f(x) = \frac{c_0}{2} + \sum_n c_n T_n(x),$$

$$c_n = \frac{2}{\pi} \int_0^\pi f(\cos(\theta)) \cos(n\theta) d\theta$$

That's all well and good, but Chebyshev polynomials have rather strange boundary conditions: $T_{2m}(-1) = T_{2m}(1) = 1$ and $T_{2m+1}(-1) = -1$, $T_{2m+1}(1) = 1$.

For the purposes of this paper, it is useful to consider how we might make the Chebyshev polynomials satisfy a Dirichlet boundary condition instead. Define the new function

$$\tau_n(x) = \begin{cases} T_{2m}(x) - T_0(x) & n \text{ even} \\ T_{2m+1}(x) - T_1(x) & n \text{ odd} \end{cases}$$

where $T_0(x) = 1$ and $T_1(x) = x$. This has the effect of making all τ_n go to 0 at both boundaries. Importantly, we can now express functions in terms of this new basis without incurring any real penalties. Boyd discusses this more in depth (pg. 29) but the conclusion is that we can treat the τ_n s as a new basis where n ranges from 2 to N .

Two Proofs

Theorem. *The Galerkin Matrix $\mathbf{L}$ is always invertible. Where $\hat{L}u = u_{xx} + V(x)u$, $u(a) = u(b) = 0$, and $L_{mn} = \langle \varphi_m, \hat{L}\varphi_n \rangle$ is the Galerkin Matrix.*

Proof. We will prove the theorem by contradiction. Assume that $\mathbf{L}$ is not invertible. Then, there must exist two vectors $\vec{c}$ and $\vec{d}$ such that

$$\mathbf{L}\vec{c} = \vec{b} \text{ and } \mathbf{L}\vec{d} = \vec{b}$$

for some $\vec{b}$ in $\mathbb{R}^n$ (a contradiction of the Invertible Matrix Theorem). We can set these equations equal to each other and find the following

$$\begin{aligned} \mathbf{L}\vec{c} &= \mathbf{L}\vec{d} \\ \mathbf{L}\vec{c} - \mathbf{L}\vec{d} &= \mathbf{0} \\ \mathbf{L}(\vec{c} - \vec{d}) &= \mathbf{0} \end{aligned}$$

Now, this could be true if $\mathbf{L}$ is the zero matrix, or if $\vec{c} = \vec{d}$. Let us now look at a component of $\mathbf{L}$

$$L_{mn} = \langle \varphi_m, \varphi_{n,xx} \rangle + \langle \varphi_m, V(x)\varphi_n \rangle$$

For the purposes of this paper, we care about looking at two specific basis functions, though the result appears to be true in general.

For sinusoidal basis functions, $\varphi_{n,xx}$ gives a constant multiplied by φ_n . This means that the term $\langle \varphi_m, \varphi_{n,xx} \rangle$ will be nonzero for all entries on the diagonal, and thus $\mathbf{L}$ is not zero.

For Chebyshev polynomials ($\varphi_n(x) = T_n(x)$), $\varphi_{n,xx}$ gives, according to Boyd (pg. 497),

$$n \frac{(n+1)T_n - U_n}{x^2 - 1}$$

where U_n are the Chebyshev polynomials of the second kind. This also contains a constant times φ_n , so $\mathbf{L}$ is not zero on at least the main diagonal.

Since both cases of basis functions that are of interest to us ended with $\mathbf{L}$ not being zero. It must be the case that $\vec{c} = \vec{d}$, but this is a contradiction. Thus, $\mathbf{L}$ must be invertible. $\square$

Theorem. *The matrix $L_{mn} = \langle \varphi_m, \hat{L}\varphi_n \rangle$ is symmetric for any operator of the form $\hat{L}u = q(x)u + c_0u + c_1u_{xx} + c_2u_{xxxx} + \dots$ where φ goes to 0 at the boundaries.*

Proof. Let us consider $\langle \varphi_m, \hat{L}\varphi_n \rangle$ in two parts. First,

$$\langle \varphi_m, q(x)\varphi_n \rangle = \int_a^b \varphi_m q(x) \varphi_n dx = \int_a^b q(x) \varphi_m \varphi_n dx = \langle q(x)\varphi_m, \varphi_n \rangle$$

which implies that the order of m and n does not affect the value of this inner product.

Second, we look at the derivative terms

$$\langle \varphi_m, c_0\varphi_n + c_1\varphi_{n,xx} + c_2\varphi_{n,xxxx} + \dots \rangle$$

Going term by term, and successively applying Integration-by-Parts (and throwing out boundary terms), we find

$$\begin{aligned} \langle \varphi_m, c_0\varphi_n \rangle &= c_0 \int_a^b \varphi_m \varphi_n dx \\ \langle \varphi_m, c_1\varphi_{n,xx} \rangle &= c_1 \int_a^b \varphi_m \varphi_{n,xx} dx \\ &= c_1 \varphi_m \varphi_{n,x} \Big|_a^b - c_1 \int_a^b \varphi_{m,x} \varphi_{n,x} dx \\ \langle \varphi_m, c_2\varphi_{n,xxxx} \rangle &= c_2 \int_a^b \varphi_m \varphi_{n,xxxx} dx \\ &= c_2 \varphi_m \varphi_{n,xxx} \Big|_a^b - c_2 \int_a^b \varphi_{m,x} \varphi_{n,xxx} dx \\ &= -c_2 \varphi_{m,x} \varphi_{n,xx} \Big|_a^b + c_2 \int_a^b \varphi_{m,xx} \varphi_{n,xx} dx \\ \langle \varphi_m, c_k\varphi_{n,x^{2k}} \rangle &= (-1)^k c_k \int_a^b \varphi_{m,x^k} \varphi_{n,x^k} dx \end{aligned}$$

where φ_{n,x^k} means that the n th φ is differentiated k times with respect to x . Note that interchanging m and n has no effect on the value of the integral for any of these terms.

Since both parts of our example operator $\hat{L}$ are symmetric, it implies that $\mathbf{L}$ will be symmetric. Further, any differential equation which has an operator of this form can be put into a weak form that will yield a symmetric matrix $\mathbf{L}$. $\square$

References

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